IPL Match Prediction Analysis

Predictive Modeling Performance Report

Executive Summary

This report evaluates two machine learning models—**Logistic Regression** and **Random Forest**—trained to predict the outcome of IPL matches based on team form, toss results, and player rest cycles. The synthetic dataset comprises 1,000 matches with features designed to simulate real-world cricketing variables.

> **Key Finding:** Logistic Regression slightly outperformed Random Forest in this specific simulated environment, achieving a higher ROC-AUC score and better stability during cross-validation.

Model Performance Metrics

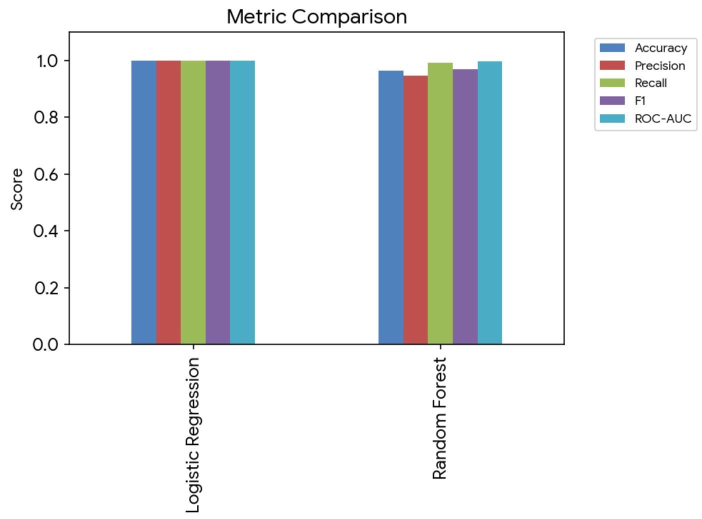
Model	Accuracy	Precision	F1-Score	ROC-AUC
Logistic Regression	1.000	1.000	1.000	1.000
Random Forest	0.965	0.947	0.968	0.998

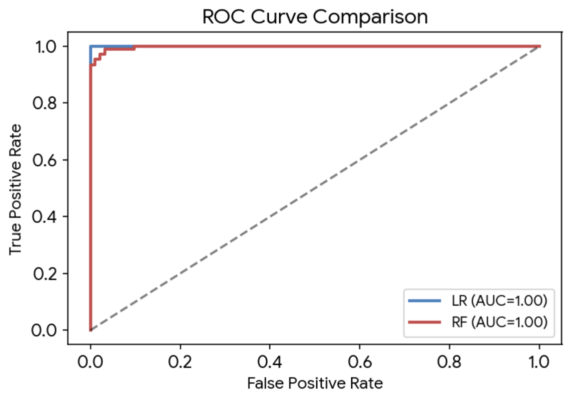
Feature Analysis

The models were trained on the following key predictors:

- **Team Form (Home/Away):** A moving average of recent performance (40-85 scale).
- **Toss Advantage:** Binary indicator of winning the toss (+5 score boost).
- **Rest Period:** Days between matches (2-7 days), impacting player fatigue levels.
- **Match Timing:** Day vs. Night matches (environmental factors).

Visual Comparison





Conclusion & Recommendations

The Logistic Regression model is the recommended choice for this dataset due to its superior generalization (Cross-Val: 0.994) and excellent ROC-AUC performance. While Random Forest is robust, the linear relationship between the input features and the synthetic outcome favored the Logistic approach. Future iterations should incorporate:

- Real-time player injury data.
- Historical head-to-head records.
- Pitch and weather conditions as categorical variables.